Digestion and Absorption of Lipids

DIGESTION OF LIPIDS

The major dietary lipids are triacyl glycerol, cholesterol and phospholipids. The average normal Asian diet contains about 20-30 g of lipids per day. Western diet generally contains two or three times more than this quantity.

Digestion in Stomach

The **lingual lipase** from the mouth enters stomach along with the food. It has an optimum pH of 2.5-5). The enzyme, therefore, continues to be active in the stomach. It acts on short chain triglycerides (SCT). SCTs are present in milk, butter and ghee. The action of lingual lipase is observed to be more significant in the newborn infants. **Gastric lipase** is acid stable, with an optimum pH around 5.4. It is secreted by chief cells, the secretion is stimulated by gastrin. Up to 30% digestion of triglycerides occurs in stomach.

Digestion in Intestines

Emulsification is a prerequisite for digestion of lipids. The lipids are dispersed into smaller droplets; surface tension is reduced; and surface area of droplets is increased. This process is favored by:

1. Bile salts (detergent action)
2. Peristalsis (mechanical mixing)
3. Phospholipids

Bile Salts are Important for Digestion of Lipids

The bile salts present in the bile (sodium glycocholate and sodium taurocholate) **lower surface tension.** They emulsify the fat droplets. The emulsification increases the surface area of the particles for enhanced activity of enzymes (Fig. 11.2).

Lipolytic Enzymes in Intestines

1. **Pancreatic lipase** with Co-lipase
2. Cholesterol esterase
3. Phospholipase A2.

The bile (pH 7.7) entering the duodenum serves to neutralise the acid chyme from the stomach and provides a pH favorable for the action of pancreatic enzymes. A list of physiologically important lipases is shown in Table 11.1.

Table 11.1. Physiologically important lipases

Lipase	Site of action	Preferred substrate	Product(s)
Lingual/acid-stable lipase	mouth, stomach	TAGs with medium/short chain FAs	FFA+DAG
Pancreatic lipase + co-lipase	small intestine	TAGs with long-chain FAs	FFA+2MAG
Intestinal lipase with bile acids	small intestine	TAGs with medium chain FAs	3 FFA+ glycerol
Phospho-lipase A_2 + bile acids	small intestine	PLs with unsat. FA on position 2	Unsat FFA lysolecithin
Lipoprotein lipase insulin (+)	capillary walls	TAGs in chylomicron or VLDL	FFA+ glycerol
Hormone sensitive lipase	adipocyte	TAG stored in adipose tissue cells	FFA+ DAG/MAG

Digestion of Triglycerides

1. **Pancreatic lipase** can easily hydrolyse the fatty acids esterified to the 1st and 3rd carbon atoms of glycerol forming 2-monoacylglycerol and two molecules of fatty acid (Fig. 11.1).
2. Then an **isomerase** shifts the ester bond from position 2 to 1. The bond in the 1st position is then hydrolysed by the **lipase** to form free glycerol and fatty acid (Fig. 11.1).
3. The major end products of the digestion of TAG are 2-MAG (78%), 1-MAG (6%), glycerol and fatty acids (14%). Thus digestion of TAG is partial (incomplete).
4. **Cholesterol** ester may be hydrolysed to free cholesterol and fatty acid. The action of **phospholipase** A2 produces lysophospholipid and a fatty acid (see Fig. 7.8).

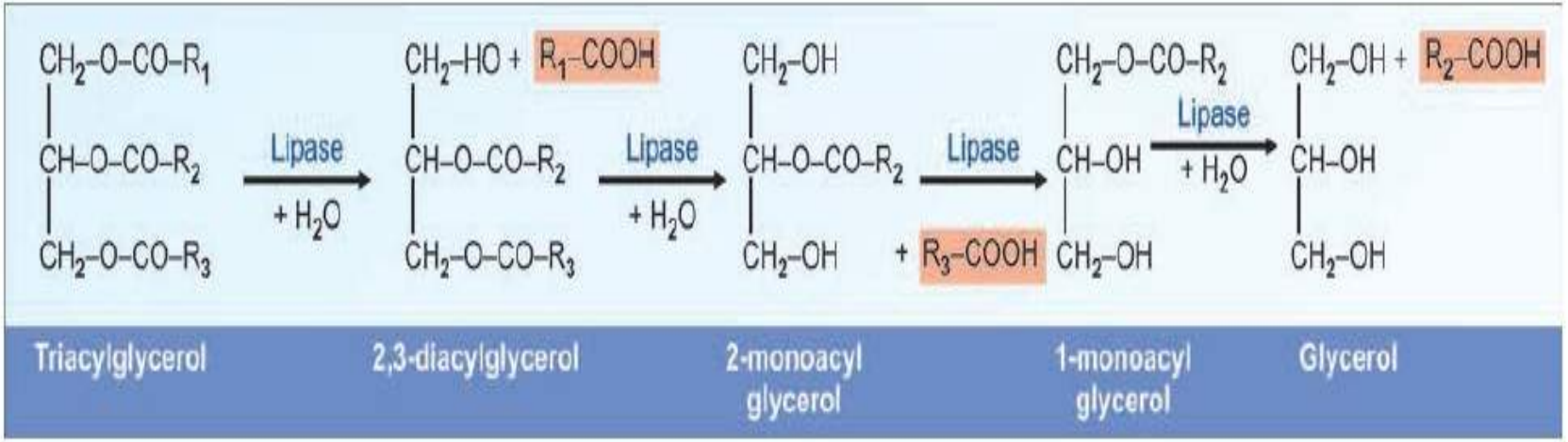


Fig. 11.1. Complete hydrolysis of triglyceride. In the intestines, generally fats are only partially hydrolysed

Co-lipase

The binding of co-lipase to the triacyl glycerol molecules at the oil water interface is obligatory for the action of lipase. The co-lipase is secreted by the pancreas as an inactive zymogen (molecular weight 11,000). It is activated by trypsin.

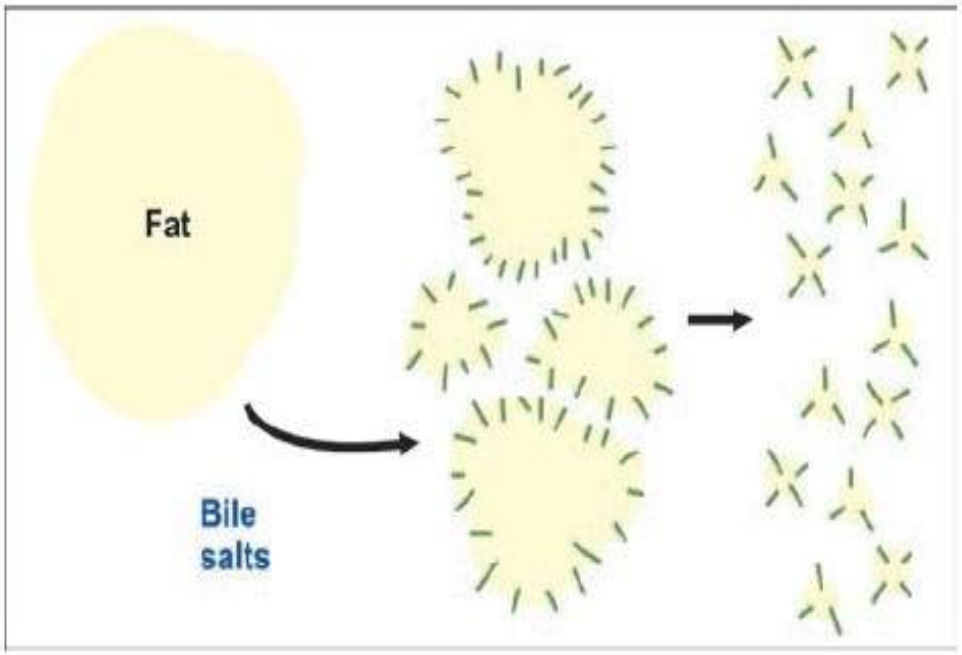


Fig. 11.2. Action of bile salts. The hydrophobic portions of bile salts intercalate into the large aggregated lipid, with the hydrophilic domains remaining at the surface. This leads to breakdown of large aggregates into smaller and smaller droplets. Thus, the surface area for action of lipase is increased

ABSORPTION OF LIPIDS

Absorption of Long Chain Fatty Acids

Long chain fatty acids (chain length more than 14 carbons) are absorbed to the lymph and not directly to the blood. The theory proposed by **Bergstrom** (Nobel Prize, 1982) has the following steps.



Bergstrom
NP, 1982
b. 1916

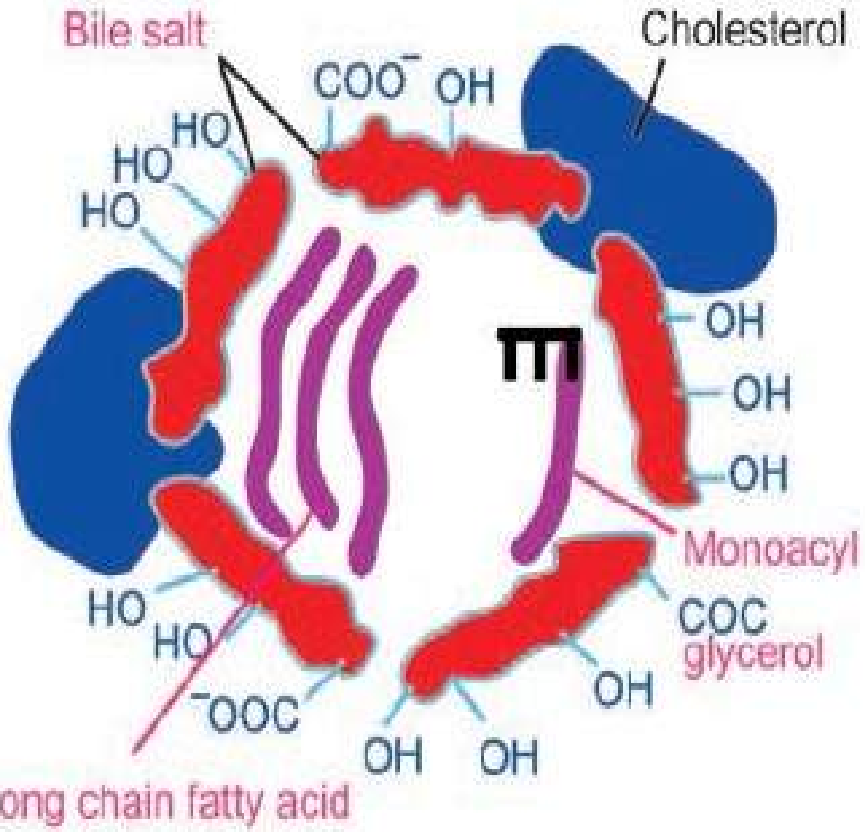


Fig. 11.3. Micellar formation

1. Mixed Micelle Formation

i. The products of digestion, namely 2-mono-glycerides, long chain fatty acids, cholesterol, phospholipids and lysophospholipids are incorporated into molecular aggregates to form **mixed micelle** (Fig. 11.3). The micelles are spherical particles with a hydrophilic exterior and hydrophobic interior core (Fig. 11.3).

Due to their detergent action, the **bile salts** help to form micellar aggregates (Fig. 11.2).

ii. Micellar formation is essential for the absorption of fat-soluble vitamins such as vitamin A, D and K.

iii. The micelles are aligned at the microvillous surface of the **jejunal mucosa**. Fatty acids, 2-MAG and other digested products passively diffuse into the mucosal cell (Fig. 11.4).

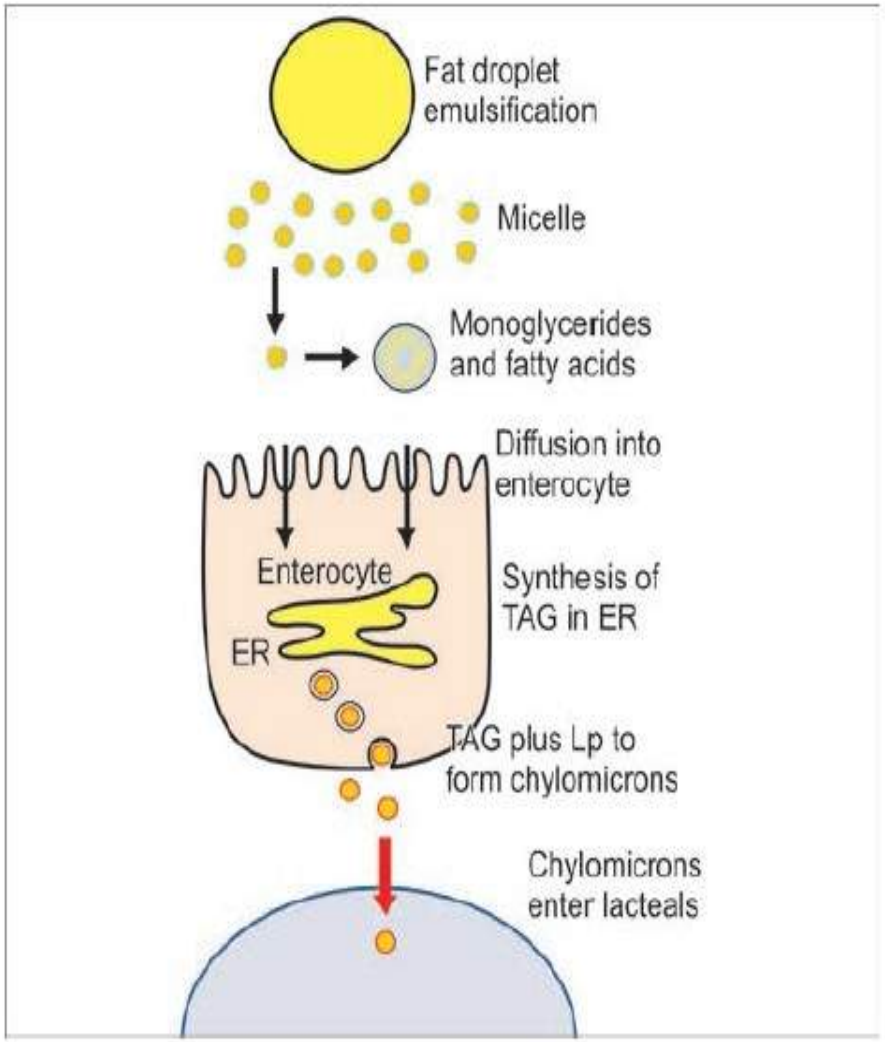


Fig. 11.4. Absorption of fat as chylomicrons. This needs the help of bile salts

The hydrophobic portions of bile salts intercalate into the large aggregated lipid, with the hydrophilic domains remaining at the surface. This leads to breakdown of large aggregates into smaller and smaller droplets. Thus the surface area for action of lipase is increased. ER = endoplasmic reticulum. TAG = triacyl glycerol

2. Enterohepatic Circulation of Bile Salts

The bile salts are left behind which are mostly reabsorbed from the ileum and returned to the liver to be re-excreted (enterohepatic circulation). About 98% of dietary lipids are normally absorbed.

3. Re-esterification Inside the Mucosal Cell

i. Once inside the intestinal mucosal cell, the long chain fatty acids are re-esterified to form triglycerides (Fig. 11.5).

ii. The fatty acids are first activated to fatty acyl CoA by the enzyme, **acyl CoA synthetase** or

thiokinase (Fig. 11.9). This needs lysis of two high energy bonds.

iii. Two such activated fatty acids react with monoacyl glycerol (MAG) to form the triglyceride. Majority of molecules follow this MAG pathway (Figs 11.4 and 11.5).

iv. Free **glycerol** absorbed from intestinal lumen directly enters into the bloodstream. So free glycerol is not available for re-esterification. But the cells can convert glucose to glycerol phosphate, and then add 3 molecules of acyl groups to synthesize TAG.

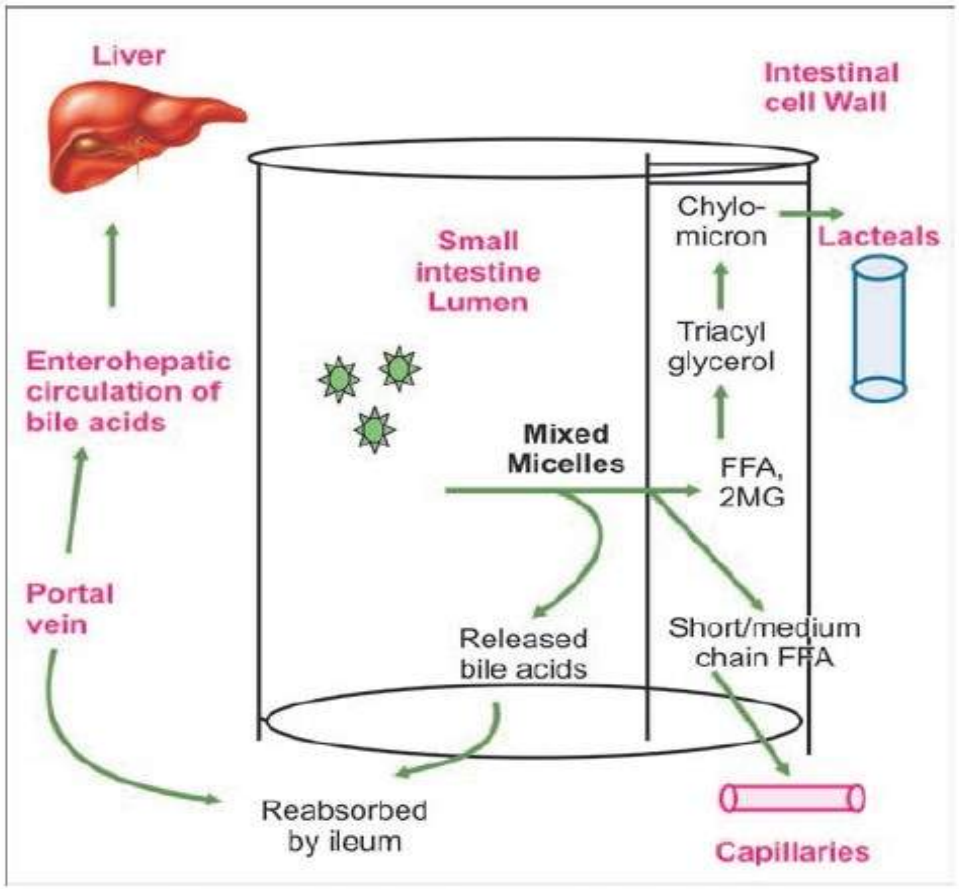


Fig. 11.5. Absorption of fatty acids

Long chain fatty acids are absorbed into the intestinal cell wall, where they are re-esterified, made into chylo-microns and enter into lymphatics. Short chain fatty acids are directly absorbed into blood capillaries. Bile acids are reabsorbed into portal vein

Chylomicrons

The TAG, cholesterol ester and phospholipid molecules along with apoproteins B48, and apo-A are incorporated into chylomicrons (Fig. 11.6). The chyle (milky fluid) from the intestinal mucosal cells loaded with chylomicrons are transported through the lacteals into the thoracic duct and then emptied into lymph circulation (Fig. 11.5). The serum may appear milky after a high fat meal (postprandial lipemia) due to the presence of chylomicrons in circulation. Normally the lipemia clears within a few hours by the uptake of chylomicrons by tissues. The major steps of lipid absorption are shown in Box 11.1.

Box 11.1. Six Steps of Lipid Absorption

1. **Minor digestion** of triacylglycerols in mouth and stomach by lingual (acid-stable) lipase.
2. **Major digestion** of all lipids in the lumen of the duodenum/jejunum by pancreatic lipolytic enzymes.
3. **Bile acid** facilitated formation of mixed micelles.
4. **Passive absorption** of the products of lipolysis from the mixed micelle into the intestinal epithelial cell.
5. **Reesterification** of 2-monoacylglycerol with free fatty acids inside the intestinal enterocyte.
6. **Assembly** of chylomicrons containing Apo B48, triacylglycerol, cholesterol esters and phospholipids **and export** from intestinal cells to the lymphatics.

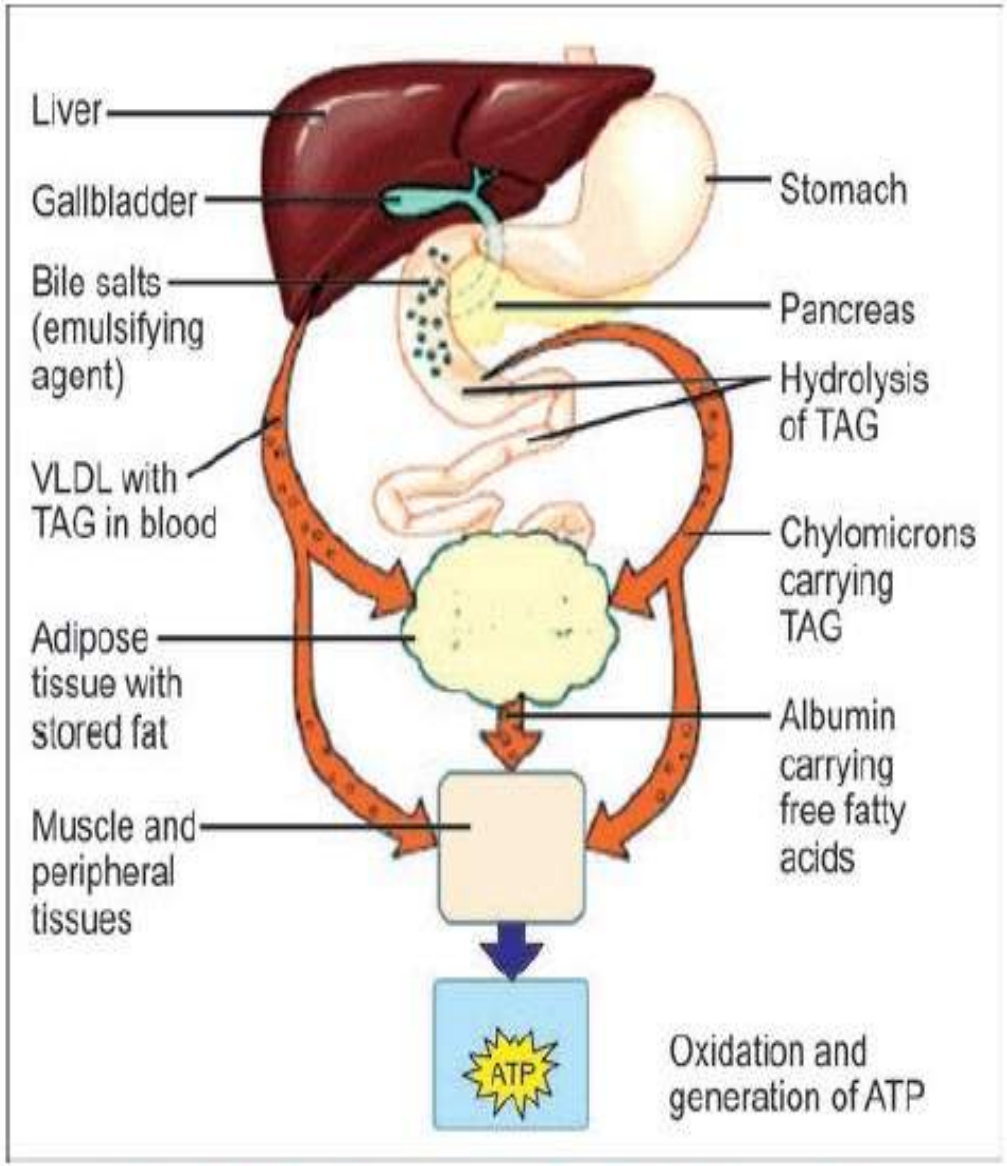


Fig. 11.6. Summary of utilization of fat

SCFA Absorption is Different

i. **Short chain** fatty acids (SCFA) (seen in milk, butter, ghee) and **medium chain** fatty acids (MCFA) (in coconut oil and mother's milk) do not need re-esterification.

ii. They can directly enter into blood vessels, then to portal vein, finally to liver where they are immediately utilised for energy (Fig. 11.5).

iii. Their absorption is rapid. They are better absorbed than long chain fatty acids. Special features in digestion and absorption of medium chain fatty acids are given in Table 13.1.

5. Abnormalities in Absorption of Lipids

1. **Defective digestion:** In steatorrhea, daily excretion of fat in feces is more than 6 g per day. (Greek word, "stear", means fat). It is due to chronic diseases of pancreas. In such cases, unsplit fat is seen in feces (Box 11.2).
2. **Defective absorption:** On the other hand, if the absorption alone is defective, most of the fat in feces may be split fat, i.e. fatty acids and monoglycerides (Box 11.2). Defective absorption may be due to diseases.

Box 11.2. Absence of Digestive Juices

1. **In pancreatic deficiency:** Steatorrhea; unsplit fat is present in stools.
2. **When bile is not available:** Absorption is defective; split fat is present in stools; defective absorption of vitamin K leads to prolonged prothrombin time.

2-A. **Celiac** disease, sprue, Crohn's disease.

2-B. **Surgical removal** of intestine.

2-C. **Obstruction of bile duct:** This may be due to gall stones, tumors of head of pancreas, enlarged lymph glands, etc. The result is deficiency of bile salts. In such cases, triglycerides with short chain and medium chain fatty acids (SCT and MCT) are digested and absorbed properly, because they do not require micellerisation for absorption. Since milk fat and coconut oil are made up of MCT, they are therapeutically useful in malabsorption syndromes.

3. **Chyluria:** There is an abnormal connection between the urinary tract and lymphatic drainage system of the intestine. Urine appears milky due to lipid droplets. **Chylothorax** can result from an abnormal connection between the pleural cavity and thoracic duct.

Fate of Chylomicrons

i. The absorbed (exogenous) triglycerides are transported in blood as **chylomicrons**. They are taken up by adipose tissue and liver.

ii. Liver synthesizes endogenous triglycerides. These are transported as **VLDL** (very low density lipoproteins) and are deposited in adipose tissue.

iii. During starvation states, triglycerides in adipose tissue are hydrolyzed to produce **free fatty acids**. In the blood, they are transported, complexed with albumin. These free fatty acids are taken up by the cells, and are then oxidised to get energy. A summary of lipid transport is shown in Figure 11.6.

Reference

- Lippincott
- Chatterjea